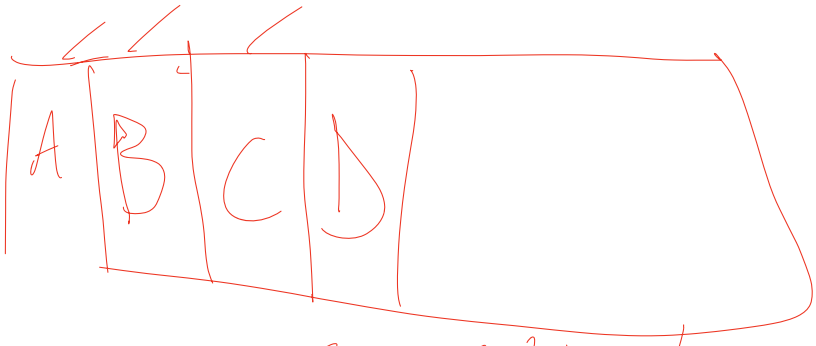




Computational complexity analysis of algorithms

```java
public static int doN(int n) {  no usages  new *
    int sum = 0;
    for (int i = 0; i < n; i++) {
        sum += i;
    }
    return sum;
}
```

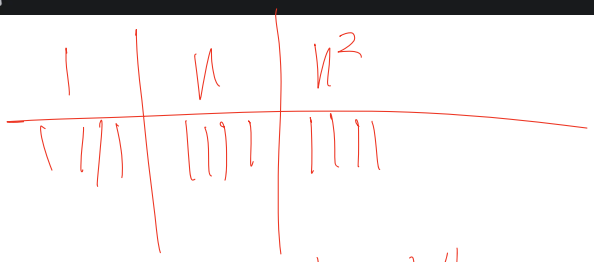
1	n
1	1

$T(n) = 4n + 4$

linear

will increase input size

```java
public static int doNTimesN(int n) {  no usages  new *
    int sum = 0;
    for (int i = 0; i < n; i++) {
        for (int k = 0; k < n; k++) {
            sum += k;
        }
    }
    return sum;
}
```



$T(n) = 4n^2 + 4n + 4$

quadratic $O(n^2)$

$T_1(n) = 4n + 4$ $O(n)$

$T_2(n) = 4n^2 + 4n + 4$ $O(n^2)$

$O(\quad)$

1. get rid of lower order terms
2. get rid of constant coefficient